Singamsetty Naga Chandrahas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.85
FIITJEE junior college	Class XII (Telangana State Board)	2024	94.90%
Vignan Bo Tree School	Class X (CBSE)	2022	94.20%

SKILLS

- **Languages:** C/C++, Java* , Python* , SQL, RISC-V, OCaml, Verilog*
- **Frameworks/Tools:** CMake, Maven, Git/Github, Linux, LaTeX, Vivado, SolidEdge, QEMU

RELEVANT COURSES

- **Mathematics:** Linear Algebra, Discrete Math, Probability & Random variables, Statistics, Calculus
- **Computer Science:** Algorithms, Operating Systems, Data Structures, Computer Architecture, C Programming, Database Management Systems, Theory Of Computation, Software Development Fundamentals

POSITIONS OF RESPONSIBILITY

- **Inter IIT Transport Team** Managed transit schedules and vehicle routing for visiting university athletes and officials during the tournament. Also coordinated on site venue setups and facilitated live match execution across sports events to ensure seamless tournament operations.
- **Teaching Assistant** for the Course Intro To Programming ID1063 (500 + students). Assisted with script checking, exam invigilation and doubt sessions.
- **Student Buddy And Mentor** at Sunshine Counselling Club, Guided and helped 20+ juniors in their first year.
- **Core member, Blockspace** Engaged in club sessions to build foundational knowledge in Web3 concepts, decentralized applications (dApps), and cryptocurrency fundamentals.

PROJECTS

Infinite Precision Arithmetic

Java / Maven

- Implemented Infinite Precision Arithmetic Engine in **Java** to handle arbitrarily Large Number computations with basic operators in Modular manner.
- Designed custom algorithms for Addition, Subtraction, Multiplication, Division for arbitrarily large integers and floating-point numbers ensuring **high precision** for complex mathematical calculations without any data loss
- Developed with Edge case Handling and Division precision upto 1000 decimals.
- Packaged the project with Maven for seamless reusability and standardized testing workflows

Custom RISC-V ISA Simulator with Cryptographic Hardware Extensions

C++ / RISC-V / Verilog / Vivado / FPGA

- Extended an in-house RISC-V ISA simulator with 12+ custom R-type/I-type instructions (GCD, powmod, isprime, invmod, ror/rol, andn/orn) to accelerate cryptographic primitives used in RSA, Diffie-Hellman, and ECC.
- Designed a modular-inverse pseudo-instruction using Fermat's Little Theorem, compressing a 30-40 line

multi-branch software routine into a single macro-instruction expanding into 7 hardware-level steps.
- Implemented and synthesized RTL modules in Verilog for custom instructions (GCD, POWMOD, BREV, ROL/ROR), validating timing and functionality on an Artix-7 FPGA at 200 MHz.
- Verified correctness of all instructions via cross-validation against independent Python and C++ reference implementations, achieving full functional consistency across isolated and integration-level test suites.

Extension of xv6 Operating System Architecture

C / Makefile

- Developed and Integrated Multi Level Feedback Queue(MLFQ) scheduler into the xv6 kernel, with global priority boosting to prevent starvation and dynamically optimize CPU utilization.
- Optimized an asynchronous virtual memory swap subsystem by implementing FCFS and SSTF disk scheduling algorithms, successfully minimizing disk head seek latency.
- Built fault-tolerant RAID-backed storage system including RAID 0,1, and 5, involving block striping for performance , drive mirroring for redundancy and distributed XOR parity mapping for recovery.

EXTRACURRICULAR

Competitive Programming Reached peak rating of 1278 in codeforces